ETAS RTA-BSW v12.7.0

RTA-SWCL Reference Guide

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1 rba_CCL

Module name	rba_CCL
Package	RTA-SWCL
AUTOSAR Version	22-11

1.1 Introduction

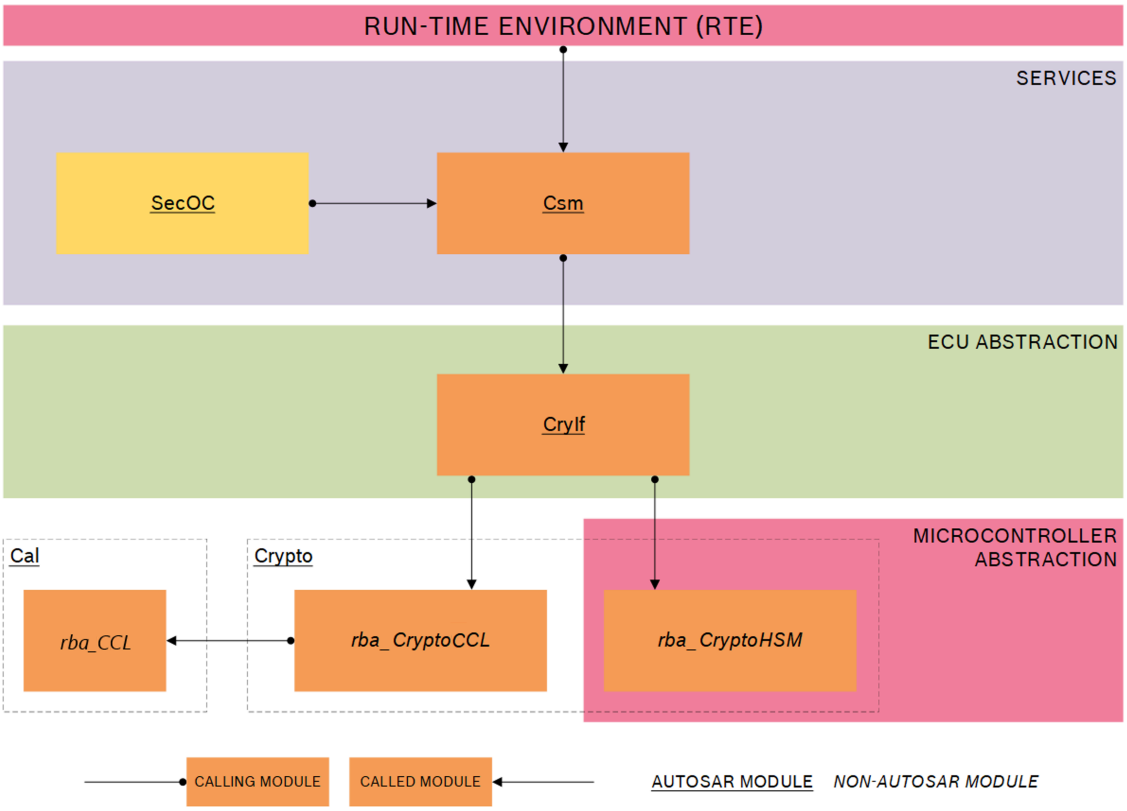

Figure 1.1.Cryptographic Module Architecture

The Cryptographic Library module (rba_CCL) provides low-level cryptographic services in the form of API functions which are called by the module rba_CryptoCCL. The API functions are abstracted as per the AUTOSAR specification for the Cal (Cryptographic Abstraction Library), however as the Cal module is not implemented in RTA-BSW, configuration of rba_CCL relies on the configuration of the Crypto Driver (Crypto) module (see AUTOSAR_SWS_CryptoDriver.pdf).

As the rba_CCL module is a library, it is not directly related to a specific layer in the AUTOSAR layered architecture. The diagram above shows the interaction between the rba_CCL module and the other modules which together provide cryptographic functionality for use in AUTOSAR projects.

This module does not support automatic configuration - it is pre-configured and must be imported into the ISOLAR-AB project workspace manually.

1.2 Supported Features

The cryptographic functions provided by the rba_CCL module are broadly defined by the following categories:

- Authenticated Encryption with Associated Data (AEAD)
- Asymmetric encryption and decryption (Asymc)
- Block cipher (Blkc)
- Block cipher modes of operations (BlckMode)
- Certificate Parsing
- Diffie-Hellman (DH) shared secret computation interface
- Deterministic Random Bit Generator (Drbg)
- Hash algorithms
- Key Derivation Functions (KDF)
- Key Wrapping
- Message Authentication Codes (MAC)
- Menezes-Qu-Vanstone common key computation and key confirmation interface
- Password Authenticated Key Exchange (Pake)
- Signature Generation and Verification
- Stream Cipher Functions

Only a subset of these features are available in order to support the overall features of the security services within RTA-BSW. See the ecy_lib_common_-config.h file.

For more information regarding the functions and their algorithms, contact ETAS.

1.3 Extensions

- No known extension parameters are defined in this version.

1.4 Limitations

This module has no known limitations.

1.5 Exclusions

- There are no known exclusions in this version.

1.6 Deviations

- There are no known deviations in this version.

1.7 API Definition

For more information regarding the functions and their algorithms, contact ETAS.

1.8 Configuration Advice

1.8.1 Configuration Parameters

As the rba_CCL module is a library, it is not configured directly using the normal mechanism of configuration parameters. Configuration of the module is done in the AUTOSAR-specified Crypto Driver module (Crypto). See AUTOSAR_SWS_CryptoDriver.pdf and the Crypto Reference Guide for more details.

1.9 Integration Advice

1.9.1 Project Integration

No specific handling necessary

1.9.2 Init Functions

No specific handling necessary.

1.9.3 DeInit Functions

No specific handling necessary

1.9.4 Scheduled Functions

There are no scheduled functions associated with the module.

1.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The Rba_CCL module does not use any exclusive OS resources.

1.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

2 Glossary

AEAD

Authenticated Encryption with Associated Data.

Asymc

Asymmetric encryption and decryption.

Blkc

Asymmetric encryption and decryption.

Cipher

A generic term for an encryption algorithm of some kind.

DRBG

Deterministic Random Bit Generator.

KDF

Key Derivation Functions.

MAC

Message Authentication Code. An algorithm used to authenticate a transmitted message.

Pake

Password Authenticated Key Exchange.